



BSc (Hons) in Information Technology
Year 3 - Semester 2 (2026)
IT3060 - Human Computer Interaction

Lab Sheet 2

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Group: 2.1

Activity 1: Organize Interview / Survey / Observation Data

1.1 Gather Raw Research Data

Data source	Type	Description	Volume
Interview transcripts	Qualitative	Transcriptions/quotes from 5 participants (3 regular users, 2 fitness trainers)	5 sessions
Survey responses	Quantitative	Type form responses on satisfaction, personalization, nutrition logging, feature interest	Piloted set and 5 pilot responses
Observation notes	Qualitative	Checklist observations from 2 trainer-led sessions on using the app usage during workout sessions	2 sessions
App analytics report	Quantitative	Onboarding drop-off, session duration and, feature usage logs	68% onboarding drop off
App store reviews	Qualitative	Public feedback reflecting rating drop from 4.6 to 3.8 stars	Sampled reviews

1.2 Clean the Data

- Removed duplicate raw data entries (eg: duplicating “nutrition logging is tedious” comment for P1 twice).
- Removed off topic remarks that did not relate to the FitFlow research goals (eg: casual small talk during interview).
- Standardized participant identifiers to P1-P5 across all sources for consistency.
- Corrected obviously erroneous typos in raw notes prior to analysis.

1.3 Categorization of Data

Category	Data Included
By source	Interviews (P1–P5), Survey, Observation checklists, App analytics, App store reviews
By user persona	Regular users (P1, P2), Lapsed users (P3), Fitness trainers (P4, P5)
Onboarding	Lapses with the onboarding process of P3; Onboarding drop-off of 68%
Workout experience	Boring workout plans (P2), Interest in AI adaptive plans (P1, P4)
Nutrition tracking	Manual food logging (P1, P5) Interest in camera-based food logging
Motivation	Loneliness and demotivation (personas), Interest in badges and reminders (P2)
Social features	Interest in community challenges and accountability (P2, P4, P5)

1.4 Storage and Tagging

- All cleaned up data was organized into a single Google Sheets workbook, consisting of Interview, Survey, Observation, and Analytics tabs.
- Miro board was used to hold tags/quotes for Activity 3 (affinity diagramming).
- Each quote was tagged with the corresponding topic and participant number in Miro.
- Sample quotes.

Sample Tagged Quotes

Participant	Quote / Observation	Tag
P1	Logging every meal takes too much time	Nutrition
P1	Need adaptive workouts depending on my schedule	Personalization
P2	Boredom with workout plans	Motivation
P3	App failed to understand my goals	Personalization
P4	AI should take into account my injuries and fitness level	Personalization
P5	Must be able to track nutrition automatically	Nutrition

Activity 2: Perform Thematic Analysis

2.1 Initial Codes

Code	Supported by	Example Evidence
Generic / repetitive workout plans	P2, P3	"Workout plans become boring"/ Generic workouts cited as main reason for lapsing.
Desire for adaptive AI plans	P1, P4	Wants workouts to fit schedule/ Wants AI to consider personal injuries and fitness levels.
Tedious nutrition logging	P1, P5	"Logging every meal takes too long"/ Food logging should be automatic
Desire for social accountability	P2, P4, P5	Interested in community challenges/ Social groups increase consistency.
Confusing onboarding	P3, App Analytics	Found onboarding confusing/ Analytics metric showing 68% onboarding abandonment.
Lack of motivation / celebration	P2	Found onboarding confusing/ Analytics metric showing 68% onboarding abandonment.
Privacy concerns around social features	P5	Privacy controls are important

2.2 Major Themes and Sub-Themes

Theme	Sub-Themes	Strength (frequency)	Supporting Evidence
Personalization Gap	Generic plans; no adaptive AI	5/5 participants	P1, P3, P4 quotes: AI workout recommendations are universally desired (5/5)
Social Isolation	No accountability partners; no community	4/5 participants	P2, P4, P5 quotes: strong demand for social and community features (4/5)
High Friction in Tracking	Manual nutrition logging is slow	4/5 participants	P1, P5 quotes: aligns with the 68% onboarding drop-off metric

Waning Motivation	No badges/celebration; boredom	4/5 participants	P2 quotes: progress tracking heavily motivates active users (4/5)
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2.3 Theme Summaries

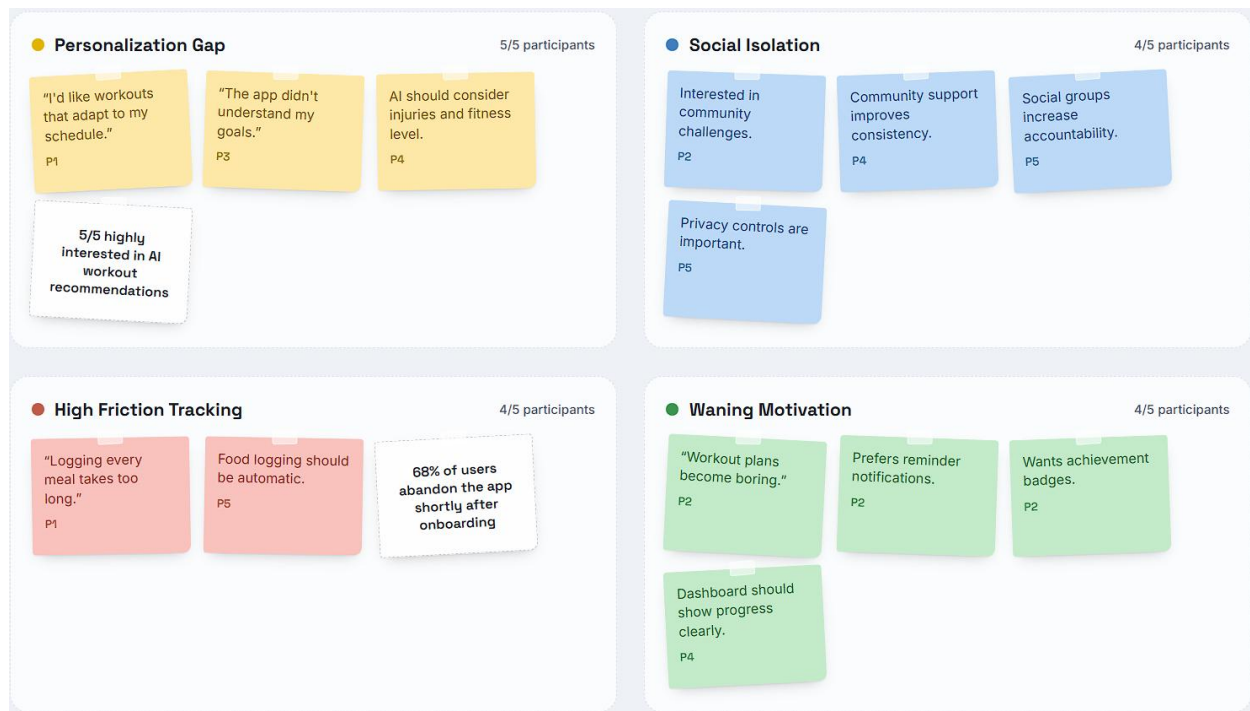
- **Personalization Gap:** Users consistently pointed out that the app's current workout plans feel generic and disconnected from their schedules, injuries, or personal fitness levels. Since all five participants expressed desire in AI adaptive workout plans, we considered this issue to be our priority.
- **Social Isolation:** Both users and trainers agreed that adding private social circles or challenges would definitely help them to be accountable and committed. However, at the same time, users warned about possible privacy breaches, indicating that any sharing features have to be completely opt-in
- **High Friction in Tracking:** Manually logging meals was listed by multiple users as one of the biggest annoyances of the app. A trainer participant separately suggested using camera-based meal logging solution.
- **Waning Motivation:** Lack of motivation components like achievement badges, clear visual progress tracking, and streaks causes users to become uninterested in further usage.

2.4 Validation

In order to validate the results, the themes were compared to the onboarding abandonment rate of 68%, which is quantified, as well as the interviews and observations carried out qualitatively. These revealed that these were not one-off comments but recurrent experiences of both regular users and fitness trainers. In addition to that, the results correlated with the main themes uncovered in the FitFlow case study, such as the personalization gap, social isolation, tracking difficulties, and decreased motivation.

Activity 3: Create an Affinity Diagram

3.1 Affinity diagram



3.2 Connections Between Clusters

- **Personalization Gap and The Waning Motivation** are related to each other since generic plans lower relevance, which makes people lose their motivation to continue.
- **The Social Isolation and Waning Motivation** are related to each other since users who do not have any accountability partners lose their motivation faster.
- **High Friction Tracking** leads to a higher rate of drop-off during onboarding, when personalization starts failing for users.

3.3 Key Findings from the Affinity Diagram

- The problem of personalization appears to be the most cross-cutting one since it concerns workout plans, nutrition and onboarding process.
- The issue of motivation cannot be solved through workout intensity only since the main reason is the absence of social mechanics.
- High friction of nutrition tracking was mentioned by all stakeholders (users and trainers independently).

Activity 4: Develop Three Personas and Empathy Maps

Persona: Alex Rivera – The Distracted Professional

"I just need a plan that fits into my chaotic day — not one more thing to plan around"

Demographics	32 years old marketing professional, works out 3-4x/week when he can manage to find time, uses FitFlow from his phone while commuting or during lunch breaks
Goals	Maintains consistency in spite of hectic work schedule and gets efficient workouts; saves time on workout planning.
Behaviors	Skips workouts if the workout plan does not work for him and gives up on long term multi week workouts.
Frustrations	Makes Alex frustrated with the inflexible approach of creating a workout plan; makes him feel that the app doesn't compensate for missed workouts.

Persona: Priya Singh – The Beginner Seeking Community

"I don't want to work out alone — I need someone to check in on me"

Demographics	27 years old, teacher, new to exercising regularly, works out 2-3 times a week, motivated by peer encouragement.
Goals	To form a regular habit; needs to find social connections that are safe and not judgmental
Behaviors	Seeks challenges and social groups on the app and is engaged more when encouraged by peers and disengaged while working out alone
Frustrations	Feels lonely and demotivated without accountability partners and unsure about connecting with others on the app

Persona: Kasun Fernando – The Lapsed User

"The app felt generic, so I stopped opening it after two weeks"

Demographics	24 years old, recently graduated from university, downloaded the app FitFlow with
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	the intention of starting a fitness journey but abandoned it
Goals	He wants an app that will know his intentions immediately and will not tolerate a tedious onboarding process
Behaviors	Tried the app briefly at the onboarding stage, felt that the workouts were too generic, stopped logging food due to the time it took and then uninstalled the app
Frustrations	There was no personalization at all, and manual logging of nutrients was just a hassle without any reason to come back

4.1 Empathy Maps

Empathy Map – Alex Rivera

Say	Think
"I need workouts that adapt to my schedule."	"If this app understood my calendar, I'd actually stick with it."
Do	Feel
Opens the app in short bursts between meetings, skips sessions that don't fit	Frustrated and guilty when missing sessions, hopeful about AI personalization
Pains	Gains
Rigid plans and no easy way to reschedule or adapt sessions on the fly	Quick adaptive plans and visible progress despite an irregular routine

Empathy Map – Priya Singh

Say	Think
"I don't want to work out alone."	"Maybe I'd stick with it if a friend was doing it with me."

Say	Think
Do	Feel
Looks for community challenges and engages more when a friend reminds her.	Lonely during solo workouts and excited by the idea of group support.
Pains	Gains
No safe, low pressure way to connect with others inside the app.	Accountability partners and positive reinforcement from a community.

Empathy Map – Kasun Fernando

Say	Think
"The app didn't understand my goals."	"Why bother logging meals if nothing about the plan is really for me?"
Do	Feel
Used the app briefly, skipped food logging, and uninstalled it after two weeks.	Indifferent and disengaged early on.
Pains	Gains
Generic workouts and tedious manual logging with no visible payoff.	A fast relevant first experience and automatic or low-effort tracking.

Activity 5: Write User Stories

	User Story	Priority	Linked Persona/Theme
1	As a busy professional, I need adaptive workouts that are generated by the AI, so I will stay consistent, regardless of my chaotic schedule	High	Alex Rivera/ Personalization Gap
2	As a beginner, I would like to have the ability to join private groups where I will feel supported and accountable for my progress	High	Priya Singh/ Social Isolation
3	As an ex-user, I would like to have onboarding process that is going to be relevant and will reflect my personal goals immediately	High	Kasun Fernando/ Personalization Gap
4	As a user, I would like to track my meals just by taking pictures of them, so that the process becomes less time-consuming	High	All personas/ High Friction Tracking
5	As a user, I would like to get some badges and see my streaks, so that I will be motivated to exercise regularly	Medium	Priya Singh/ Waning Motivation
6	As a user, I would like to get my missed workouts automatically rescheduled, so that I wouldn't break my plan because of one missed day.	High	Alex Rivera/ Personalization Gap

7	As a fitness trainer, I need to customize AI-generated workout plans for clients to benefit from professional advice.	Medium	Trainer stakeholders/ Personalization Gap
8	As a user, I need to restrict the visibility of my activity in social features in order to feel secure about my progress sharing	High	Priya Singh/ Social Isolation
9	As a user, I need a visualization of my progress on the dashboard so that I would feel accomplished.	Medium	Alex Rivera/ Waning Motivation
10	As a new user, I need an intuitive and short onboarding flow to keep me in the application and avoid quitting without exploring all of its benefits.	High	Kasun Fernando/ High Friction Tracking
11	As a user, I need reminders for logging my workouts and meals based on my schedule so that I would not forget to do it.	Medium	Priya Singh/ Waning Motivation
12	As a user, I need to join challenges along with my friends in order to stay motivated with the help of competition.	Medium	Priya Singh/ Social Isolation

Activity 6: Create User Journey Maps

6.1 Current-State User Journey

Stage	Actions	Thoughts	Emotions	Pain Points/ Opportunities
Awareness	Downloads FitFlow after seeing an ad or recommendation	"Let's see if this helps me get fit."	Curious, hopeful	Opportunity: set expectations early about personalization
Onboarding	Completes generic sign up and initial fitness questionnaire	"This feels like every other app."	Neutral to mildly frustrated	Pain: 68% of users abandon shortly after onboarding
First Workout	Follows a pre-set generic plan	"This isn't really for me."	Disappointed	Pain: no adaptive personalization
Nutrition Logging	Manually enters meals	"This takes too long."	Annoyed, fatigued	Pain: high friction in tracking
Ongoing Use	Uses app sporadically, skips days, works out alone	"I don't feel like anyone notices if I stop."	Lonely, unmotivated	Pain: social isolation, waning motivation
Churn	Stops opening the app and eventually uninstalls	"It wasn't worth the effort."	Indifferent	Opportunity: win-back via improved AI and social features

6.2 Future-State (Ideal) User Journey

Stage	Actions	Thoughts	Emotions	Supporting Feature
Awareness	Downloads FitFlow after seeing AI coaching messaging	"This sounds like it actually adapts to me."	Curious, optimistic	Targeted marketing on AI coaching
Onboarding	Completes short, adaptive onboarding that sets an initial AI plan	"It already understands what I need."	Engaged, reassured	AI workout engine and streamlined onboarding flow

First Workout	Follows an AI routine tailored specifically to current time constraints	"This actually fits my day."	Motivated, confident	AI "Daily Flow" recommendation card
Nutrition Logging	Snaps a quick photo for instant food recognition	"That was fast."	Satisfied, relieved	Camera based nutrition logging
Ongoing Use	Joins a private social circle and earns streaks/badges	"My friends and I keep each other going."	Connected, motivated	Private social circles and progress dashboard
Retention	Continues using app and considers premium subscription	"This app actually gets me."	Loyal, satisfied	Personalized progress insights and community challenges

Activity 7: How Might We (HMW) Statements

Statement	Description
HMW 1 (Personalization)	How might we make the AI-generated fitness and nutrition plans feel authentic to every individual's schedule, fitness level, and goals?
HMW 2 (Social Engagement)	How might we design social functions in order to encourage users' accountability and community feeling, but not expose or intimidate them?
HMW 3 (Nutrition Tracking)	How might we make nutrition logging process to such an extent that it ceases to be a deterrent to using the application?
HMW 4 (User Onboarding & Motivation)	How might we design onboarding and progress feedback so that new users feel understood immediately and stay motivated over time?

Activity 8: Functional and Non-Functional Requirements with Traceability

8.1 Functional Requirements

ID	Requirement	Priority	Source
FR1	The system shall generate an AI-powered workout plan that adapts to the user's schedule, fitness level, and stated goals.	High	HMW1; Story 1, 6; Alex Rivera persona
FR2	The system shall allow users to create and join private social circles with configurable visibility/privacy settings.	High	HMW2; Story 2, 8; Priya Singh persona
FR3	The system shall provide camera-based (computer vision) food recognition for nutrition logging.	High	HMW3; Story 4; High Friction Tracking theme
FR4	The system shall display a personalized progress dashboard showing streaks, achievements, and workout history.	Medium	HMW4; Story 5, 9; Waning Motivation theme
FR5	The system shall provide a short, adaptive onboarding flow that produces an initial AI plan within the first session.	High	HMW4; Story 10; Kasun Fernando persona
FR6	The system shall automatically suggest	Medium	Story 6; Alex Rivera persona

	a rescheduled session when a planned workout is missed.		
FR7	The system shall allow fitness trainers to view and adjust AI-generated plans for their clients.	Medium	Story 7; Trainer stakeholder group
FR8	The system shall send configurable reminder notifications for workouts and meal logging.	Medium	Story 11; Waning Motivation theme
FR9	The system shall support community challenges that groups of users can join together.	Medium	Story 12; Social Isolation theme

8.2 Non-Functional Requirements

ID	Requirement	Priority	Source
NFR1	The mobile app shall load core screens (dashboard, workout, logging) within 2 seconds on a standard connection.	High	Case study non-functional goals; retention concerns
NFR2	The app shall support offline logging of workouts and meals, syncing automatically once connectivity resumes.	Medium	Case study non-functional goals
NFR3	The app's interface shall comply with WCAG 2.1 AA accessibility standards.	High	Case study non-functional goals; final accessibility audits
NFR4	All personal and health data shall be encrypted at rest and in transit, in	High	Consent forms; data security/privacy findings (P5)

	compliance with GDPR and CCPA.		
NFR5	AI personalization models shall run on-device where feasible to minimize transmission of sensitive user data.	Medium	Technology stack justification (TensorFlow Lite)
NFR6	The system shall maintain 99.5% uptime for social and notification features.	Medium	Backend scalability requirements (Firebase, Node.js/Express)

8.3 Traceability Matrix

Req. ID	Related Theme	Related Persona(s)	Related User Story	Related HMW
FR1	Personalization Gap	Alex Rivera	Story 1, 6	HMW1
FR2	Social Isolation	Priya Singh	Story 2, 8	HMW2
FR3	High Friction Tracking	All personas	Story 4	HMW3
FR4	Waning Motivation	Priya Singh, Alex Rivera	Story 5, 9	HMW4
FR5	Personalization Gap	Kasun Fernando	Story 10	HMW4
FR6	Personalization Gap	Alex Rivera	Story 6	HMW1
FR7	Personalization Gap	Trainer stakeholders	Story 7	HMW1
FR8	Waning Motivation	Priya Singh	Story 11	HMW4
FR9	Social Isolation	Priya Singh	Story 12	HMW2
NFR1	High Friction Tracking	All personas	—	HMW3
NFR2	High Friction Tracking	All personas	Story 4	HMW3
NFR3	(Accessibility compliance)	Kasun Fernando	Story 10	HMW4
NFR4	Social Isolation (privacy)	Priya Singh	Story 8	HMW2
NFR5	Personalization Gap	All personas	Story 1	HMW1

NFR6	Social Isolation	Priya Singh	Story 2, 12	HMW2
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8.4 Review Notes

- **Clarity:** Every requirement is formulated as a single testable claim with an actor (the system/app) and expected behavior.
- **Feasibility:** All requirements are consistent with the justified technology stack (React Native, Node.js/Express, Firebase, TensorFlow Lite), which means that no requirement is beyond the team's technical capabilities.
- **Completeness:** All major topics (Personalization Gap, Social Isolation, High-Friction Tracking, Fading Motivation) and HMWs are represented in functional requirements, and privacy/performance issues voiced by participants are included into non-functional requirements.